

Does *how you phrase* an instruction decide how much an LLM's output wobbles?

A crossed-design benchmark measuring prompt sensitivity in summarization, replicated across two model families — Llama-3.1-8B & Qwen2.5-7B.

Task	Scale
Summarization (CNN/DailyMail)	2 models × 8,970 generations

Design	Date
30 articles × 50 seeds × ~6 paraphrases	Aug 2026

The one-sentence takeaway

Which rhetorical *dimension* an instruction uses — question-form, theme-isolation, output-format, and so on — drives prompt sensitivity **far more** than the coarse Pool A/B split we originally hypothesised. This holds strongly in Llama-3.1-8B and **replicates in direction** in Qwen2.5-7B, with the most- and least-sensitive dimension classes **identical** across both models.

~11x

Dimension explains more sensitivity variance than Pool split (Llama: 19.3% vs 1.7%)

0.70

Cross-model dimension-ranking agreement (Spearman, $p = 0.005$)

0.87

Split-half replication within the study (mean over 300 splits)

17,940

Total generations analysed across both models

What problem are we studying?

Prompt sensitivity

LLMs give **inconsistent outputs** when a prompt is reworded but its **meaning is preserved**. That fragility undermines reproducibility and trust. We ask: **is this fragility random, or is it structured by the style of instruction?**

Our clean definition

Sensitivity = spread of a cell's outputs. **Quality = mean**. We keep the two strictly separate so a prompt can't look "sensitive" just because it's low-quality.

A **cell** = one (article, seed) → its ~6 meaning-equivalent paraphrase outputs.

Main finding — Llama-3.1-8B

Dimension is the primary axis. Instruction dimension explains **19.3%** of sensitivity variance vs just **1.7%** for the Pool A/B split (~11x). Mixed-effects test `sensitivity ~ dimension + (1|seed) + (1|article)` : $\chi^2 = 30.8$, **p = 0.0036**. The Pool A/B split is **not** significant (Mann-Whitney p = 0.16).

Honest qualifier: the effect is **strong in Llama** but **weaker/borderline in Qwen** (assumption-free permutation p = 0.071). Only the **coarse** ranking replicates — the exact 14-way order is model-specific.

Per-dimension sensitivity ranking

Dimension	Sens.	
Meta-reflection *1 seed	0.620	most
Question Form	0.614	most
Theme Isolation	0.611	most
Analytical Framing	0.561	
Role Based	0.520	
Persona / Tone	0.490	
Audience Perspective	0.487	
Emphasis Focus	0.485	
Completeness Framing	0.483	
Tone Register	0.480	
Lexical Filter	0.468	
Direct Command	0.446	least
Output Format	0.358	least
Constraint Based	0.352	least

* Meta-reflection rests on 1 seed — treat as a single-prompt observation, not a stable estimate.

Robustness validation (Llama) — every internal check passed

Check	Question	Result	
Different scorers (x3 configs)	Is the ranking scorer-dependent?	Spearman 0.94	stable
Quality confound	Is sensitivity just low quality?	residualized 0.94	passed
Length confound	Just an output-length effect?	corr -0.07; adj. 0.97	passed
Split-half replication	Does it replicate within the study?	mean Spearman 0.87 (300 splits)	passed
Permutation test	Significant with no normality assumption?	p = 0.022	passed
Paraphrase equivalence	Do paraphrases truly mean the same?	bidir. NLI 0.98 / 0.97, 100% passed	passed

The **assumption-free permutation p = 0.022** is the most defensible significance value — it makes no normality assumption that the rank-normalized composite would violate.

Cross-model replication — Qwen2.5-7B

Identical grid and pipeline; only the generation model changed. This is the headline result for the mentor: **the structure is not a Llama artifact**.

What replicates

Metric	Result
Dimension-ranking Spearman	0.70 (p=.005)
Seed-ranking Spearman (50 prompts)	0.74 (p=1e-9)
SMS-drift anchor cross-check	0.82
Bottom-3 dimension overlap	Jaccard 1.00
Top-3 dimension overlap	Jaccard 0.50

✓ Question Form

✓ Theme Isolation

most sensitive in both.

Constraint

Direct Command

Output Format

least sensitive in both — identical set.

The critical caveat (we report it honestly)

The 0.70 replication is in **direction/ranking**, not **strength**. Under strict assumption-free tests Qwen's dimension effect does **not** clear significance:

- Qwen **permutation p = 0.071** (borderline); seed-Kruskal p = 0.156.
- Significant only in the parametric mixed model (p = 0.019).
- Llama, by contrast, **passes** the permutation test (p = 0.022).
- Excluding the ~4.5% degenerate/refusal cells did **not** rescue it (0.071 → 0.066) — the weakness is **intrinsic**.

So effect **strength is model-dependent**: strong in Llama, borderline in Qwen.

Per-dimension, side by side (sorted by Llama rank)

Dimension	Llama sens (rank)	Qwen sens (rank)	Agreement
Meta-reflection *1 seed	0.620 (1)	0.498 (7)	moved
Question Form	0.614 (2)	0.545 (2)	stable
Theme Isolation	0.611 (3)	0.646 (1)	stable
Analytical Framing	0.561 (4)	0.535 (4)	stable
Role Based	0.520 (5)	0.463 (10)	moved
Persona / Tone	0.490 (6)	0.464 (9)	
Audience Perspective	0.487 (7)	0.510 (6)	
Emphasis Focus	0.485 (8)	0.521 (5)	
Completeness Framing	0.483 (9)	0.496 (8)	
Tone Register	0.480 (10)	0.543 (3)	moved
Lexical Filter	0.468 (11)	0.445 (11)	stable
Direct Command	0.446 (12)	0.428 (13)	stable
Output Format	0.358 (13)	0.441 (12)	stable
Constraint Based	0.352 (14)	0.398 (14)	stable

Which model is more sensitive? (absolute measures)

The rank-normalized composite is relative within a run, so we use the run-independent absolute spreads on the same 1,500 cells.

What drifts across paraphrases	Llama	Qwen	More sensitive
Meaning (SMS drift)	0.0941	0.0953	≈ tie (p=.10)
Wording (ROUGE-L spread)	0.0009	0.0011	Qwen (p=6e-10)
Correctness spread	0.0041	0.0046	Qwen (p=4e-6)
BERTScore spread	0.0032	0.0038	Qwen (p=2e-7)
Faithfulness spread	0.0399	0.0383	Llama (n.s.)

Qwen is slightly more sensitive overall — but the edge is in *wording*, not *meaning* (meaning-drift is a tie). So Qwen is "moodier on average but less organized by dimension," while Llama is calmer overall but more clearly dimension-driven. **"More sensitive" and "more dimension-explained" are different axes.**

Methodology in brief

Design

- **Crossed grid:** every seed applied to the same 30 CNN/DM articles (instance-fixed, prompt-varied) — POSIX / ProSA / PromptBench convention.
- **Decoding:** greedy (temperature 0), to isolate the prompt effect from sampling noise.
- **Paraphrase equivalence** validated by **bidirectional NLI entailment**, not just similarity.

Sensitivity composite (v2), rank-normalized within a run

$$0.25 \cdot \text{SMS-drift} + 0.20 \cdot \text{ROUGE-L-var} + 0.15 \cdot \text{CS-cv} + 0.15 \cdot \text{faith-cv} + 0.125 \cdot \text{PPL-var} + 0.125 \cdot \text{PC-stab-cv}$$

Spread of bounded quality scores is level-normalized (coefficient of variation) to de-confound spread from quality level; lexical drift (ROUGE-L) keeps the score from being purely embedding-based.

Scorers — deterministic & SOTA (no LLM-judge, on purpose)

SMS drift

Sentence-BERT — meaning drift

MiniCheck-DeBERTa-v3-large

SOTA faithfulness / fact-checking (EMNLP 2024)

BERTScore (rescaled)

Correctness vs gold summary

Teacher-forced PPL / BF

Model-confidence spread

An LLM-judge was deliberately avoided — its own stochasticity would contaminate a *variance* benchmark.

Tech stack

Models & generation

Llama-3.1-8B-Instruct

Qwen2.5-7B-Instruct

vLLM (inference)

HuggingFace Transformers

PyTorch

H100 GPU

Scoring & metrics

Sentence-BERT (all-mpnet)

MiniCheck-DeBERTa-v3-large

BERTScore

ROUGE-score

NLI entailment

Analysis & stats

NumPy / Pandas

SciPy

statsmodels (mixed-effects)

Permutation / bootstrap

Split-half replication

Data

CNN/DailyMail

HF Datasets

JSONL pipeline

Statistical tests used

Mixed-effects LRT (χ^2)

Permutation test

Mann-Whitney / Kruskal

Spearman ρ

η^2 variance decomposition

Reproducibility

Git-tracked runs

Fixed seeds

Deterministic scorers

compare_models.py

Honest limitations & what's next

Limitations we're not hiding

- **Effect strength is model-dependent** — strong in Llama, borderline in Qwen.
- Only the **coarse** pattern replicates; the exact 14-way order is model-specific.
- **Thin dimensions** — Meta-reflection = 1 seed; several have only 3.
- **Narrow slice** — two 7–8B models, one task, one dataset; no 70B, no non-summarization task.
- **Post-hoc pivot** — the pre-registered Pool A/B hypothesis failed; dimension emerged from the data.
- No human validation of the drift metric yet (face validity only).

Recommended next steps

- **A third model (highest value)** — resolves whether Qwen's weakness is an outlier or the effect is genuinely fragile. Mistral-7B (new family) or Llama-3.1-70B (scale contrast).
- **Balance thin dimensions** — add seeds so no dimension rests on a single prompt.
- **~50-pair human validation** of the drift metric.
- **Format/surface perturbations** as a second axis, or scope the paper explicitly to semantic-paraphrase sensitivity.

Defensible claim for the write-up: Across two model families, instruction **dimension** drives prompt sensitivity far more than the Pool split (both mixed-model $p < 0.02$, ~11–18× more variance), and the dimension rankings agree significantly (Spearman 0.70, $p = 0.005$) with identical most/least-sensitive classes. Question-form & theme-isolation phrasings are reliably the most sensitive; format & constraint phrasings the least. Effect **strength is model-dependent** and the fine-grained ranking is model-specific — best described as a **robust coarse pattern**, not an exact cross-model law.